

ECC006 — Web Design

Homework Assignment #7

Part 1: Multiple Choice Questions

Question 1

How would you link to the named fragment #jobs on the page employ.html from the home page of the site?

- a. `<a href="employ.html#jobs">Jobs</a>`
- b. `<a name="employ.html#jobs">Jobs</a>`
- c. `<a link="employ.html#jobs">Jobs</a>`
- d. `<a href="#jobs">Jobs</a>`

Answer: a. `<a href="employ.html#jobs">Jobs</a>`

To link to a named fragment on a different page, the href attribute must contain both the filename and the fragment identifier separated by #. Option d. uses only #jobs, which only works when linking within the same page. Options b. and c. use invalid attributes that do not exist in anchor tag syntax.

Question 2

Which pseudo-element can be used to generate content that precedes an element?

- a. `:after`
- b. `:before`
- c. `:content`
- d. `:first-line`

Answer: b. `:before`

The `:before` pseudo-element inserts generated content immediately before the selected element. The `:after` pseudo-element does the opposite and inserts content after the element. The `:content` and `:first-line` pseudo-elements serve entirely different purposes and cannot generate content that precedes an element.

Question 3

Which of the following is a mobile web design best practice?

- a. Configure a multiple-column page layout.
- b. Avoid using lists to organize information.
- c. Configure a single-column page layout.
- d. Embed text in images wherever possible.

Answer: c. Configure a single-column page layout.

Small screens cannot comfortably display multiple columns side by side without requiring the user to scroll horizontally or zoom out. A single-column layout keeps all content within the visible width of the screen and makes scrolling natural and easy. Embedding text in images is discouraged because the text cannot scale properly and is unreadable by assistive technologies.

Lists are actually recommended, not avoided, as they help users scan information quickly on mobile devices.

Part 2: Find the Error (Question 4)

The page is intended to display the navigation area on the right side of the browser window. Two changes must be made to the CSS to achieve this.

Error 1 — Incorrect float direction

The nav element is set to `float: left`, which places the navigation on the left side of the page. This must be changed to `float: right` so that the navigation appears on the right.

```
Before: nav { float: left; width: 120px; }  
After:  nav { float: right; width: 120px; }
```

Error 2 — Incorrect padding on main

The main element has a padding of `20px 150px 20px 20px`, which reserves 150px of space on the right side. Since the navigation is now floating right, the large padding value must move to the left side to prevent the main content from overlapping the navigation.

```
Before: main { padding: 20px 150px 20px 20px; }  
After:  main { padding: 20px 20px 20px 150px; }
```

Part 3: Web Research Summary

Accessibility and Mobile Web Design: Areas of Overlap and Shared Best Practices

The W3C document Web Content Accessibility and Mobile Web, available at <http://www.w3.org/WAI/mobile>, shows that designing for users with disabilities and designing for mobile devices share a significant number of common goals and technical solutions. Web developers who understand this overlap can build websites that serve a much wider audience without duplicating their effort.

Alternative Text for Images

Both accessibility guidelines and mobile best practices require descriptive alt attributes on all images. Screen reader software depends on alt text to describe images to visually impaired users. Mobile users who disable image loading on slow connections also rely on alt text as the only description of an image. Writing meaningful alt attributes therefore addresses both audiences at once.

Logical Heading Structure

Proper use of HTML heading elements from h1 through h6 benefits both groups of users. Screen readers allow users to navigate a page by jumping between headings, making a clear and consistent hierarchy essential. On small mobile screens, well-structured headings also allow users to scan a page quickly without reading every word. Developers should use headings based on document structure rather than for visual styling purposes.

Single-Column Layouts

Complex multi-column layouts create problems for screen reader users because content is read in the order it appears in the HTML source, which may not match the visual layout on screen. The same layouts are also problematic on narrow mobile screens, where they require horizontal

scrolling. A single-column, linearly structured layout solves both issues simultaneously, making content readable and logically ordered.

Avoiding Text Embedded in Images

Embedding text within image files is discouraged by both accessibility and mobile guidelines. Screen readers cannot extract or read text that is part of an image. On mobile devices, image-based text does not scale when users increase their font size or when the screen resolution changes, resulting in blurry or illegible content. Using real HTML text styled with CSS avoids both problems entirely.

Keyboard and Touch Navigation

Accessible websites must be fully operable using only a keyboard, which is essential for users with motor disabilities who cannot use a mouse. This aligns closely with mobile touch navigation, which requires interactive elements to be large enough and spaced far enough apart to be tapped accurately with a finger. Both standards lead developers toward clearly defined, well-spaced buttons, links, and form fields.

Concise and Well-Organized Content

Screen reader users experience content one element at a time, making dense or poorly organized text frustrating to navigate. Mobile users face a similar challenge on small screens where long blocks of text require excessive scrolling. Both audiences benefit from short paragraphs, descriptive link text, and the use of properly structured lists to group related information.

Conclusion

The overlap between web accessibility and mobile design best practices reflects the fact that both fields address the same underlying challenge: delivering content clearly under constrained conditions. Developers who build with accessibility in mind naturally produce websites that also perform well on mobile devices. Adopting a unified approach to accessibility and mobile compatibility leads to more inclusive, maintainable, and effective web design for all users.